

help collect energy. The mouse's body interacts with surrounding magnetic fields, helping focus energy like a lens from the transmitter to the receiver in the implant.

The device is cylindrical in shape, is about 2mm wide and 3mm long, making it about 100 times smaller and lighter than previous devices created to achieve this.

Gas sensor can detect internal air levels in the body during sleep

Scientists at VTT research, a technical research firm based out of Finland, have developed a tiny gas sensor that can be connected with a smartphone or tablet to conduct gas measurements inside the human body. For example, a mobile healthcare app can be used along with the gas sensor to measure the quality of sleep with great precision by gauging CO₂ quantities in your body.

The sensor uses adjustable optical filters known as Fabry-Pérot interferometers, which have been previously used for applications based on spectroscopy, such as environmental monitoring via drones or nano-satellites, various hyperspectral cameras, early detection of skin cancer and fuel analysis for minimising emission rate.



The smartphone app shows internal air levels

The gas sensor channels light through the sample that needs to be analysed. By measuring the penetration level at various light wavelengths, composition of the sample is determined. For example, CO₂ is identified based on its strong absorption of light at a wavelength of 4.2µm. It is also possible to differentiate other gases or substances in the sample at the same time by determining the spectrum of their absorption peaks at various infrared wavelengths.

Battery-free cameras to estimate object's position

Researchers have revealed that a network of energy-harvesting sensor nodes equipped with onboard cameras can determine each object's pose and location using optical cues. This exercise could be used to monitor the condition of infrastructure like bridges, industrial equipment and home security monitoring, among others.

It could also enable networks of thousands of sensors to operate without batteries or external power, with minimal maintenance. Such networks can be part of the Internet of Things, where objects can

BINAY PowerLED

Inspection Hand Lamp

BOX Design **LINEAR Design**

BINAY PowerLED Inspection Hand Lamps utilize a light source comprised of multiple high power white LEDs

- Very low power consumption of 10 watts or less
- Long life (over 50,000 burning hours), covered under 3-year warranty
- Unbreakable shock-proof and vibration-resistant construction; cool light output with no heat radiation
- Glare-free light output, focusing only on the work area

Binay Opto Electronics Pvt. Ltd.

27, PANDITIA TERRACE, Calcutta 700 029, India
 Telephone: +91 (33) 4006 9875, 2475 0392, 2475 0030
 email: info@binayled.com, http://www.binayled.com

POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983

SIENNA ECAD
Design . Enhance . Verify

**YOUR PARTNER IN PCB DESIGN
ENGINEERING SERVICES
& PCB PROTOTYPES
FOR OVER 18 YEARS**

**MORE THAN
10000
COMPLEX DESIGNS
GERBERED**

- **PCB layout Design**
- **Analysis**
 - Signal Integrity • Power Integrity
 - Thermal • EMI/EMC
- **DFX Analysis Services**
 - DfA • DfM
- **Prototypes**
 - Quick Turn Around PCB Prototypes

Right from PCB designs to Manufacturing & Assembly Services

Cont : 080 30410700 | E-mail : sales@siennaecad.com
www.siennaecad.com